2026-04-30

HydraR: Agentic Infrastructure for Reproducible Scientific Research

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Executive Summary

Large Language Models (LLMs) are catalyzing a paradigm shift in scientific computing, fundamentally transforming how research is performed. This transformation is particularly evident in the field of bioinformatics, where LLMs have evolved beyond simple chatbots into autonomous "agents" that can plan and execute complex tasks that previously demanded computational expertise, and powerful multi-agent systems where a team of AI agents can autonomously collaborate with one another to tackle multi-faceted problems. However, the lack of a robust framework to support the development and use of agentic systems in the R ecosystem poses a significant risk, whereby the ephemeral nature of agent reasoning and potentially destructive filesystem modifications, threaten to undermine the integrity of R-based research. Because of this, there is a critical need for traceable, stateful and secure agent orchestration. In this project, we address this need by building on HydraR, an existing R package dedicated to the scientific reproducibility of agentic systems. Specifically, this project has two distinct objectives. First, we aim to develop a comprehensive memory and state management system for automated retries of failed tasks, execution caching and state retrieval for reducing redundant computations, and validation of agent outputs for workflow resilience. Second, we aim to develop a robust environment management system that integrates with filesystem and runtime isolation tools such as Git worktree and Docker, to ensure that agents can execute safely without risking the integrity of sensitive research environments. By developing these tools, HydraR will become a flexible, agent-agnostic framework compatible with any orchestration package, thereby empowering computational biologists to build reproducible AI-driven workflows that uphold the rigor and integrity of R-based scientific research.

Budget

We request **$14,400 USD** in funding for an expert software developer to work on this project over a 7-week period (approximately 245 hours).

Signatories

Project Team

- **Chi Nam (Ignatius) Pang - Co-Primary Investigator**. Senior Bioinformatician at Macquarie University / Australian Proteome Analysis Facility (APAF). Expert in R package development, proteomics pipelines, and reproducibility standards.
- **Aidan Tay - Co-Primary Investigator**. Bioinformatician at APAF. Architect of the HydraR R6 DAG engine, Git worktree isolation logic, and the custom state-hashing persistence Layer.

Contributors

- **APAF Bioinformatics Team**: Provided the testing ground for initial prototypes in high-throughput proteomics and metabolomics research.
- **Antigravity AI**: Agentic assistant used during the rapid prototyping and documentation hardening phase of the project (as disclosed in `agents.md`).

Community Supporters

- **Hanna Szafranska**: PhD student at University of Cambridge.
- **Nathalie Nataren**: PhD Student at Adelaide University.
- **Tyrone Chen**: Bioinformatician at Peter MacCallum Cancer Centre.
- **Calum Walsh**: Bioinformatician at University of Melbourne.
- **Johan Gustafsson**: Research Community Engagement Lead at Australian BioCommons.

Consulted

- **rOpenSci Editorial Board**: Initial discussions regarding HydraR's role in the AI-R ecosystem.
- **Macquarie University**: Consulted on the impact of agentic orchestration for national-scale omics infrastructure.
- **Henrik Bengtsson**: Consulted on HydraR package design and package infrastructure.
- **Jeroen Ooms**: Consulted on HydraR package design and package infrastructure.

The Problem

Large Language Models (LLMs) are catalyzing a paradigm shift in scientific computing, fundamentally transforming how research is performed. This transformation is particularly evident in the field of bioinformatics, where LLMs have evolved beyond simple chatbots into autonomous "agents" that can plan and execute complex tasks previously requiring computational expertise, and into powerful multi-agent systems where a team of AI agents can autonomously collaborate to tackle multi-faceted problems. By democratizing access to these sophisticated multi-agent systems, this automation not only accelerates biological discovery but also empowers the broader scientific community to harness high-performance workflows once restricted to deep domain specialists.

Yet despite offering many great benefits, **the current R ecosystem lacks the necessary framework for traceable, stateful, and secure agent orchestration**. R packages like `ellmer` and `mall` provide excellent interfaces but fail to record agent decisions, making the process ephemeral and untraceable. Meanwhile, advanced frameworks such as `LangGraph` and `AutoGen` in the Python ecosystem lack the necessary features for auditable agent orchestration, rendering them unsuitable for

R-based research via Python-based bridges (i.e., `reticulate`). Moreover, granting autonomous agents unchecked access to research environments can inadvertently lead to unwanted file modifications, deletions, or corruption of the working directory. While packages such as `LLMagentR` and `puppeteeR` enable multi-agent orchestration and state management, `puppeteeR` lacks restart from checkpoint capabilities and both `LLMagentR` and `puppeteeR` do not implement environment isolation for agents. Thus, to safeguard the integrity of R-based research, a robust framework must exist to support the development and reproducible use of agentic systems.

To address this critical gap in the R ecosystem, this project will build on HydraR, an existing R package dedicated to the scientific reproducibility of agentic systems. Specifically, we aim to develop:

1. A comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions.
2. A robust environment management system for isolating agents from the host environment.

By delivering these systems, **we will equip computational biologists with the necessary tools to develop production-grade agentic systems**, such as automated retries of failed tasks, execution caching for reducing redundant computations, and safe execution of agents without risking the integrity of sensitive research environments. Crucially, **developing these tools will transform HydraR into a flexible, agent-agnostic framework compatible with any orchestration package**, and therefore an infrastructure hub for the next generation of agentic tools. By serving as the central foundation for traceable, stateful, and secure agent orchestration, HydraR will empower computational biologists and R developers alike to adopt agentic systems with confidence, enabling them to build reproducible AI-driven workflows that uphold the rigorous standards of scientific research.

The Proposal

Overview

This project addresses the critical need for traceable, stateful, and secure agent orchestration in R by building upon HydraR (https://github.com/APAF-bioinformatics/HydraR), a package dedicated to the reproducibility of agentic systems. Specifically, **we aim to develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions, and a robust environment management system for isolating agents from the host environment**. These systems will equip computational biologists with the tools to develop production-grade agentic systems, by enabling automated retries, reducing redundant computations, and safe execution within sensitive research environments. By developing these tools, HydraR will become a flexible, agent-agnostic framework compatible with any orchestration package, transforming HydraR into an infrastructure hub for the next generation of agentic tools.

Details

To bridge the gap in the R ecosystem and deliver a robust framework for traceable, stateful and secure agent orchestration, this project will leverage the existing functionality of HydraR. Specifically, we aim to address this critical need through two objectives:

Objective 1: Memory and State Management

Aim To develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions.

Rationale Agent memory is often ephemeral and untraceable, making it difficult to capture and audit the entire "journey" of an agentic workflow. While HydraR uses DuckDB for continuous checkpointing and verifiable records, **it currently lacks specialized features required for production-grade bioinformatic pipelines**, such as automated retries, execution caching, and validation of agent outputs for workflow resilience.

Approach To develop a comprehensive memory and state management system, we will introduce several enhancements to HydraR:

- **Restart Checkpointing**: We will implement a system to automatically serialize agents, environment, local variables, and their current reasoning state. In the event of a failure (e.g., API timeout, resource exhaustion, or logic error), HydraR can reconstitute the exact state of the R session, allowing for automated retries without restarting the entire pipeline.
- **Execution Caching and State Retrieval**: We will implement a content-addressable storage (CAS) model for all agent-generated artifacts. By hashing output files and R objects, and linking them to specific reasoning paths, HydraR can maintain a complete audit trail of how each artifact was generated, allowing the system to "skip" redundant tasks and "resume" workflows from any saved state.
- **Agent Validation and Recovery**: We will implement a system to automatically validate agent outputs. If the output of an agent fails to validate (e.g., malformed code or incorrect data schema), HydraR can diagnose and rectify the error, ensuring the main research flow continues safely and autonomously.

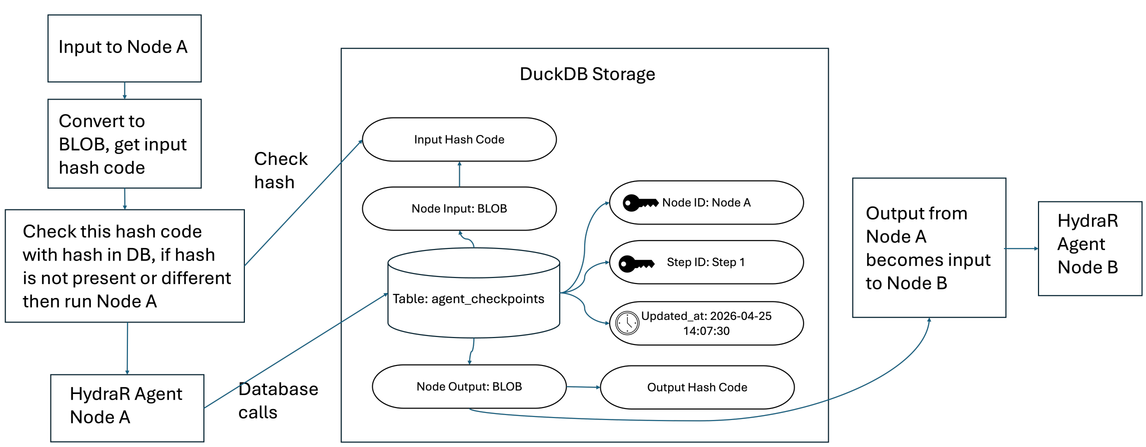

Figure 1: Memory and State Management

Objective 2: Environment Management

Aim To develop a robust environment management system for isolating agents from the host environment.

Rationale Granting autonomous agents unchecked agent access to research environments risks unintended file modifications, deletions, or corruption of the working directory. Unrestricted access to the working directory also makes it difficult for parallelised agents to work on the same task without overwriting each other's work, hindering scalability and interoperability. Currently, **HydraR does not enforce filesystem and runtime isolation, lacking the capabilities to safely and securely orchestrate agentic workflows** and impeding the adoption of agentic systems in R.

Approach To develop a robust environment management system, we will introduce several new features to HydraR:

- **Git Worktrees**: We will implement a Git worktree layer to prevent agents from modifying the filesystem at the same time. By assigning every task to a unique worktree, HydraR can capture and track all file changes made by the agent, thereby providing an auditable record of every file modification.
- **Docker Integration**: We will implement a Docker layer to isolate agent execution from the host environment. By utilizing ephemeral containers that can be deployed and shut down on demand, HydraR will ensure that every agent operates in a secure environment. This approach will also guarantee workflow reproducibility by providing a consistent, standardized environment for every execution.
- **Singularity and Apptainer for HPC**: We will provide native support for Singularity and Apptainer to accommodate bioinformatic workflows hosted institutional High-Performance Computing (HPC) clusters where Docker may be restricted. However, we will optimize this feature for standard HPC structures (e.g., SLURM integration) to ensure that the agentic infrastructure complies with security and resource policies.
- **Persistent State Tracking via Mounted Worktrees**: To bridge the gap between container volatility and workflow statefulness, we will manage Git worktree logic within persistent Docker mounts. By mounting the worktree directory from the host to the container, HydraR will ensure that the agent's filesystem state is preserved across container restarts. Combined with our restart checkpointing, this will allow an agent to resume from the exact filesystem state it was in before a crash, while maintaining a versioned history of its progress via Git.

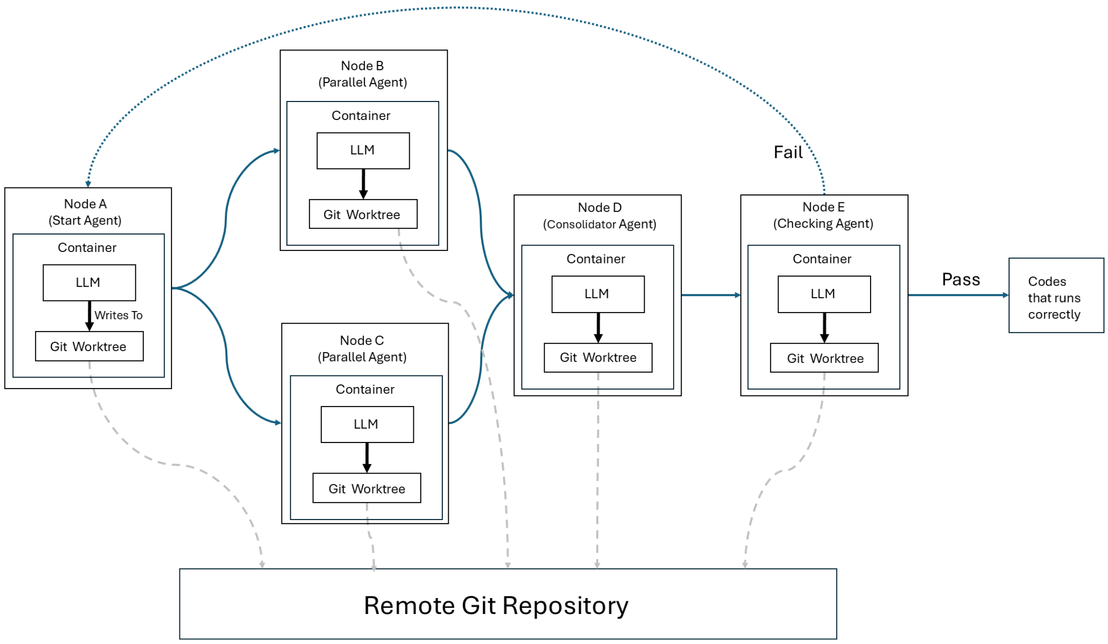

Figure 2: Environment Management

Timeline

Start-up Phase

- **Reporting Framework**: Establish a bi-weekly internal review process to track milestone progress.

Technical Delivery

The proposed work will be delivered over a 7-week period and involve approximately 245 hours of expert developer time. The work is structured into three phases:

- **Phase 1: Memory and State Management (Weeks 1-3)**:
  - Development of automated restart checkpointing system.
  - Development of execution caching and state retrieval system.
  - Development of agent validation and recovery system.
- **Phase 2: Environment Management (Weeks 4-6)**:

- Integration with Git worktree.
- Integration with Docker integration.
- Compatibility for Singularity/Apptainer.

- **Phase 3: Validation & Community Management (Week 7)**:
  - Finalization of Agents and Skills Management tools into lifecycle modules.
  - Documentation, bug fixing, and release processing.

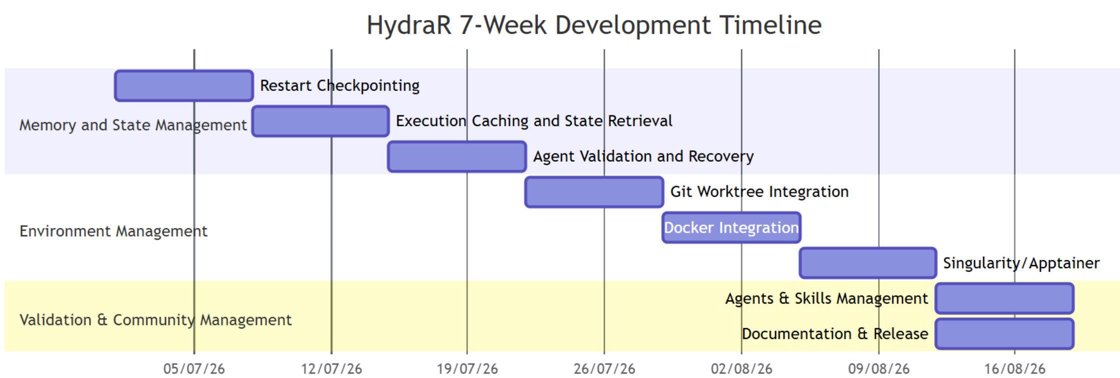

Figure 3: Timeline

Other Aspects

- **Hosting**: Code will be hosted on GitHub under the `APAF-bioinformatics` organization.
- **Conference**: We will attend and present HydraR at relevant conferences to engage with the bioinformatics community.
- **Social media**: We will utilize social media to engage with the bioinformatics community.
- **Publicity**: We will share quarterly updates on the R Consortium blog.
- **Community Engagement**: An early version of the tool is already undergoing the rOpenSci peer-review process and will transition to a community-maintained model.

Budget & Funding Plan

- Total requested budget: **$14,400 USD**.
- This grant will solely support funding for Co-CI Tay. Co-CI Pang is currently employed by the Australian Proteome Analysis Facility (APAF) at Macquarie University and will contribute in an advisory capacity and as a user beta-tester.
- Macquarie University will provide support by providing access to High-Performance Computing (HPC) resources for testing and development.

Milestones Budget

Description	R ConsortiumCash (USD)
Milestone 1: Memory and State Management	
Co-CI Tay (1.0 FTE)	$5,760
Milestone 2: Environment Management	
Co-CI Tay (1.0 FTE)	$5,760
Milestone 3: Validation & Community Management	
Co-CI Tay (1.0 FTE)	$2,880
Total	**$14,400**

Success

The success of the HydraR infrastructure project will be measured by its adoption, technical stability, and contribution to the R ecosystem's AI readiness.

Success Metrics

- **Package Adoption**: At least 3 existing R/Bioconductor packages demonstrating "Skill Node" integration via the HydraR manifest protocol.
- **Reliability Benchmark**: A published reproducible benchmark dataset showing a >20% improvement in agent task completion rates when using HydraR state management vs. stateless alternatives.
- **Community Engagement**: Successful completion of the rOpenSci peer-review process and transition to a community-maintained model. We will seek to demonstrate the tool via online webinars or community user groups hosted by the Australian BioCommons (https://www.biocommons.org.au/), local workshops hosted by APAF and Macquarie University and as submitted abstract at relevant bioinformatics and proteomics conferences.
- **CRAN/Bioconductor Presence**: Maintain the 0-error/warning/note status across all major R versions.

Definition of Done

The project will be considered successful upon the delivery of:

- A stable CRAN-ready (or Bioconductor-ready) R package named `HydraR`.
- A fully functional memory and state management system that can autonomously retry failed tasks, skip previously executed tasks, or validate agent outputs.
- A fully functional environment memory system that integrates with Git worktree, Docker and Singularity/Apptainer.
- Comprehensive documentation, including at least three "reproducible case studies" (vignettes) showcasing traceable agent orchestration.

Measuring success

We will use the following markers to track our progress:

- **Week 2**: Completion of the `duckdb` backend and successful hashing of R objects in a session.
- **Week 4**: Completion of agent validation and integration with Git worktree.
- **Week 6**: Completion of integration with Docker and compatibility for Singularity/Apptainer
- **Week 7**: Successful community beta-test with at least 5 bioinformatics researchers and final project report and publication of the demonstration blog post on the R Consortium website.

Future work

HydraR is backed by **APAF Bioinformatics** (at Macquarie University), ensuring long-term institutional support and alignment with national-scale bioinformatics infrastructure (NCRIS).

- **Commercial & Institutional Support**: Ongoing maintenance is supported by funding of APAF bioinformatics staff until end of 2028. Co-CI Pang will continue to maintain the package after the grant period as part of his role at APAF.
- **Open Source Community**: By providing an interoperable manifest protocol, we encourage a "plugin" ecosystem where community members can contribute drivers and skill nodes without needing deep knowledge of the core DAG engine, ensuring the framework outlives its original funding period.

HydraR is designed to be the foundation for an "Autopoietic Agentic Ecosystem" in R, with autopoietic defined as the capacity of the system to maintain and reproduce itself. Future development directions include:

- **Decentralized Orchestration**: Extending the infrastructure to support peer-to-peer agent coordination across distributed R nodes.
- **Self-maintaining agents**: Agents that can autonomously create or remove agent nodes and to dynamically reconfigure their connections in response to the outputs of other agents.
- **Advanced Containerization**: Tighter integration with Docker/Singularity for "Secure Execution Enclaves" where agents can test code in isolated environments.
- **Agentic Benchmarking**: A standardized suite for benchmarking LLM performance on bioinformatics-specific R tasks using the HydraR traceability layer.